# PHISHING URL DETECTOR

## Aim

Develop Machine learning based web site for Phishing URL Detection




## Abstract

Phishing is a major cyber threat where attackers use fake websites to steal personal information. This project presents a machine learning–based web tool that detects whether a given URL is safe or phishing. The system extracts important features from the URL—such as length, suspicious characters, HTTPS usage, and keywords—and uses a trained ML model to classify it. The web app provides instant results and explanations, helping users identify harmful links and improve online safety.

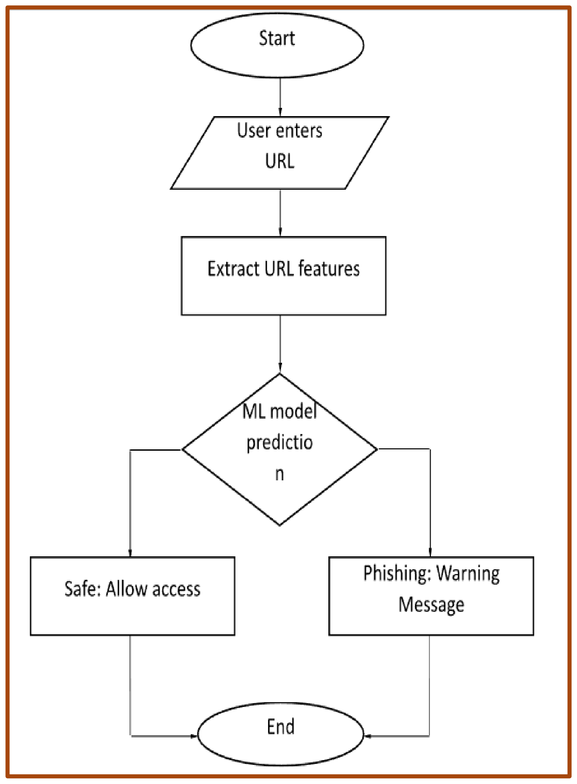
## Problem Statement

Phishing websites look similar to legitimate ones and trick users into sharing passwords, credit card details, or personal information.

People cannot manually detect suspicious URLs easily — there is a need for a simple tool that instantly analyzes and flags dangerous links.

## Proposed Solution

- Build a web app where:
- User pastes a URL
- System extracts features (length, '@', IP address present, domain age, etc.)
- ML model or rule-based system classifies URL as Safe / Suspicious / Phishing
- App shows detailed explanation (why it's dangerous)
- This helps users detect risky links before opening them.




## ADVANTAGES

- **Reduces Cyber Fraud:** Prevents password theft, banking fraud, and data loss.
- **Real-Time Detection:** Instantly analyzes and blocks harmful URLs.2
- **High Accuracy:** Machine learning improves detection over manual or rule-based methods.
- **User-Friendly:** Easy for anyone to spot phishing links.
- **Scalable:** Works across browsers, apps, email clients, and enterprise systems.

## OUTCOME

- Early detection of harmful phishing URLs
- Reduces cyber-fraud and information theft
- Helps users browse safely
- Can be integrated into browser extensions, mobile apps, or email scanners

## CONCLUSION

The Phishing URL Detector effectively identifies malicious links using machine learning. It contributes to online safety, reduces cyber-attacks, and provides a reliable solution to protect users from phishing scams. The system can be expanded into browser plugins or integrated into enterprise-level security infrastructure.

**"Think Before You Click!"**